SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BCSE302P - DATABASE SYSTEMS LAB

Semester : Winter Semester 25-26 **Faculty : Dr.S.Geetha**

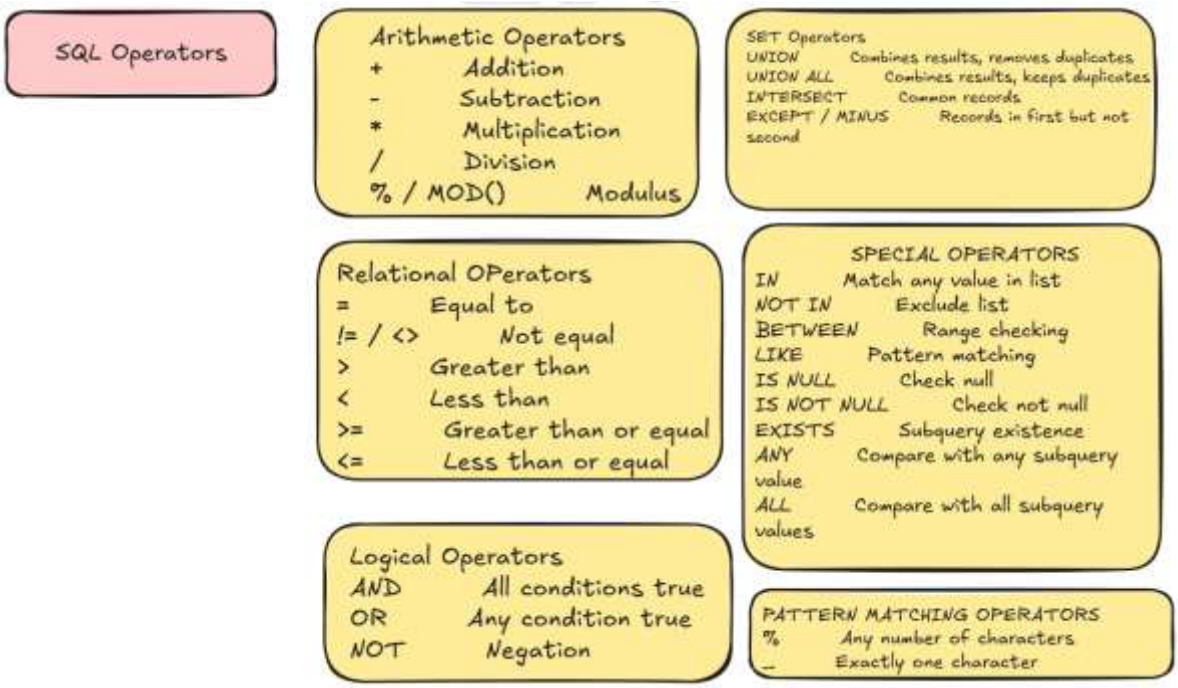
EXERCISE 4

Title: SQL Operators.

1. OBJECTIVE

- To apply SQL operators in queries
- To use aggregate functions for data analysis
- To practice SQL built-in functions (string, numeric, date, and null-handling)

2. SQL OPERATORS



STUDENT TABLE

student_id	student_name	dept	age
101	Arun	CSE	20
102	Banu	ECE	21
103	Charan	CSE	19
104	Divya	MECH	22
105	Eshwar	CSE	21
106	Farah	CIVIL	20

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

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COURSE TABLE

course_id	course_name	credits
201	Database Management Systems	4
202	Operating Systems	3
203	Computer Networks	3
204	Data Structures	4
205	Software Engineering	3

ENROLLMENT TABLE

enroll_id	student_id	course_id	marks	semester
1	101	201	78	3
2	101	204	85	3
3	102	201	68	3
4	102	203	72	3
5	103	204	90	2
6	103	202	75	2
7	104	205	60	4
8	105	201	88	3
9	105	203	82	3
10	106	205	55	4

A. ARITHMETIC OPERATORS

- Used in numeric calculations.

Operator	Meaning	Example
+	Addition	salary + bonus
-	Subtraction	salary - tax
*	Multiplication	salary * 12
/	Division	salary / 30
% / MOD()	Modulus	marks % 2

1. Display the marks of all students after adding 5 grace marks (without updating the table).
   SELECT enroll_id, marks, marks + 5 AS updated_marks
   FROM ENROLLMENT;

2. Add 5 grace marks to all students enrolled in Semester 3.
   UPDATE ENROLLMENT
   SET marks = marks + 5
   WHERE semester = 3;

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3. Display the marks of all students after deducting 10 marks as penalty (without updating the table).
```
SELECT enroll_id, marks, marks - 10 AS reduced_marks
FROM ENROLLMENT;
```

4. Deduct 10 marks from students who scored less than 60.
```
UPDATE ENROLLMENT
SET marks = marks - 10
WHERE marks < 60;
```

5. Display course credits after doubling the credit value (without updating the table).
```
SELECT course_id, credits, credits * 2 AS doubled_credits
FROM COURSE;
```

6. Double the credits of all 4-credit courses.
```
UPDATE COURSE
SET credits = credits * 2
WHERE credits = 4;
```

7. Display half of the marks scored by students (for analysis purpose only).
```
SELECT enroll_id, marks, marks / 2 AS half_marks
FROM ENROLLMENT;
```

8. Reduce marks to half for students enrolled in semester 4 (disciplinary action).
```
UPDATE ENROLLMENT
SET marks = marks / 2
WHERE semester = 4;
```

9. Display the remainder when marks are divided by 2 to identify even and odd marks.
```
SELECT enroll_id, marks, marks % 2 AS remainder
FROM ENROLLMENT;
```

10. Give 1 extra mark to students who currently have odd marks.
```
UPDATE ENROLLMENT
SET marks = marks + 1
WHERE marks % 2 = 1;
```

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B. RELATIONAL (COMPARISON) OPERATORS

- Used in WHERE, HAVING, JOIN.

Operator	Meaning
=	Equal to
!= / <>	Not equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

11. Display details of students belonging to CSE department.

```
SELECT * FROM STUDENT
WHERE dept = 'CSE';
```

12. Display students who are not from ECE department.

```
SELECT * FROM STUDENT
WHERE dept <> 'ECE';
```

13. Display enrollment details of students who scored more than 80 marks.

```
SELECT * FROM ENROLLMENT
WHERE marks > 80;
```

14. Display students who scored less than 60 marks.

```
SELECT * FROM ENROLLMENT
WHERE marks < 60;
```

15. Display students whose age is 20 years or above.

```
SELECT * FROM STUDENT
WHERE age >= 20;
```

16. Display courses having credits less than or equal to 3.

```
SELECT * FROM COURSE
WHERE credits <= 3;
```

C. LOGICAL OPERATORS

Operator	Meaning
AND	All conditions true
OR	Any condition true
NOT	Negation

17. Display the details of students who belong to the CSE department and whose age is greater than 20.

```
SELECT *
FROM STUDENT
WHERE dept = 'CSE' AND age > 20;
```

18. Display the details of students who belong to CSE OR ECE department.

```
SELECT *
FROM STUDENT
WHERE dept = 'CSE' OR dept = 'ECE';
```

19. Display the details of students who do not belong to the MECH department.

```
SELECT *
FROM STUDENT
WHERE NOT dept = 'MECH';
```

D. SET OPERATORS

Used to combine results of two SELECT queries.

Operator	Description
UNION	Combines results, removes duplicates
UNION ALL	Combines results, keeps duplicates
INTERSECT	Common records
EXCEPT / MINUS	Records in first but not second

20. Display student IDs from CSE department and ECE department (without duplicates).

```
SELECT student_id FROM STUDENT WHERE dept = 'CSE'
UNION
SELECT student_id FROM STUDENT WHERE dept = 'ECE';
```

21. Display student IDs from CSE and ECE departments including duplicates.

```
SELECT student_id FROM STUDENT WHERE dept = 'CSE'
UNION ALL
SELECT student_id FROM STUDENT WHERE dept = 'ECE';
```

22. Display student IDs who are enrolled in semester 3 AND scored more than 75 marks.

```
SELECT student_id FROM ENROLLMENT WHERE semester = 3
INTERSECT
SELECT student_id FROM ENROLLMENT WHERE marks > 75;
```

23. Display students who are not enrolled in any course.

```
SELECT student_id FROM STUDENT
EXCEPT
SELECT student_id FROM ENROLLMENT;
```

E. SPECIAL OPERATORS

Operator	Purpose
IN	Match any value in list
NOT IN	Exclude list
BETWEEN	Range checking
LIKE	Pattern matching
IS NULL	Check null
IS NOT NULL	Check not null
EXISTS	Subquery existence
ANY	Compare with any subquery value
ALL	Compare with all subquery values

24. Display the details of students who belong to CSE or ECE departments.

```
SELECT *
FROM STUDENT
WHERE dept IN ('CSE', 'ECE');
```

25. Display the details of students who are NOT from CSE or ECE departments.

```
SELECT *
FROM STUDENT
WHERE dept NOT IN ('CSE', 'ECE');
```

26. Display the enrollment details of students who scored marks between 70 and 85.

```
SELECT *
FROM ENROLLMENT
WHERE marks BETWEEN 70 AND 85;
```

27. Display the details of students whose names start with the letter 'A'.

```
SELECT *
```

```
FROM STUDENT
WHERE student_name LIKE 'A%';
```

28. Display the enrollment details of students whose marks are not yet entered.

```
SELECT *
FROM ENROLLMENT
WHERE marks IS NULL;
```

29. Update the marks of student 106 for course 205 to NULL (marks not evaluated) and display the updated record.

```
UPDATE ENROLLMENT
SET marks = NULL
WHERE student_id = 106 AND course_id = 205;

SELECT *
FROM ENROLLMENT
WHERE marks IS NULL;
```

30. Display the enrollment details where marks are available (not NULL).

```
SELECT *
FROM ENROLLMENT
WHERE marks IS NOT NULL;
```

31. Display all students if there exists at least one student enrolled in the CSE department.

```
SELECT *
FROM STUDENT S
WHERE EXISTS (
   SELECT 1
   FROM STUDENT
   WHERE dept = 'CSE'
);
```

32. Display students who have scored more marks than ANY student enrolled in course 205.

```
SELECT DISTINCT student_id
FROM ENROLLMENT
WHERE marks > ANY (
   SELECT marks
   FROM ENROLLMENT
   WHERE course_id = 205
);
```

33. Display students who have scored more marks than ALL students enrolled in course 205.

```
SELECT DISTINCT student_id
FROM ENROLLMENT
WHERE marks > ALL (
   SELECT marks
   FROM ENROLLMENT
   WHERE course_id = 205
);
```

F. PATTERN MATCHING OPERATORS

Symbol	Meaning
%	Any number of characters
_	Exactly one character

34. Display the details of students whose names start with the letter 'A'.

```
SELECT *
FROM STUDENT
WHERE student_name LIKE 'A%';
```

35. List students whose names end with the letter 'a'.

```
SELECT *
FROM STUDENT
WHERE student_name LIKE '%a';
```

36. Display students whose names contain the letter 'a' anywhere.

```
SELECT *
FROM STUDENT
WHERE student_name LIKE '%a%';
```

37. Display students whose names have exactly 5 characters.

```
SELECT *
FROM STUDENT
WHERE student_name LIKE 'D____';
```

38. Display students whose second letter is 'a'.

```
SELECT *
FROM STUDENT
WHERE student_name LIKE '_a%';
```

39. Display the details of students whose names start with 'C' and end with 'n'.

```
SELECT * FROM STUDENT
WHERE student_name LIKE 'C%n';
```

3. EXPECTED OUTCOMES

After executing this experiment, the student will be able to:

- Understand the use of arithmetic operators for computed results without modifying table data.
- Apply relational operators to filter records based on conditions.
- Use logical operators to combine multiple conditions in a query.
- Implement special operators such as IN, BETWEEN, IS NULL, and EXISTS.
- Use pattern matching operators (LIKE, %, _) to retrieve records based on textual patterns.
- Analyze query results and interpret how operators affect row selection.

4. EXPERIMENT QUESTIONS

- What is the difference between WHERE and HAVING clauses?
- How does the % operator differ from the _ operator in SQL?
- Why are aggregate functions not allowed in the WHERE clause?
- Explain the difference between IN and EXISTS with an example.
- What is the significance of IS NULL and why can't = be used with NULL values?.

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5. **PRACTICE DATABASE –** Create these tables with suitable constraints and indicated records.

TABLE 1: EMPLOYEE

emp_id	emp_name	dept	salary	age
1	Anand	IT	45000	28
2	Bhavya	HR	38000	32
3	Charan	IT	52000	26
4	Divya	FINANCE	60000	35
5	Ezhil	HR	40000	29
6	Farooq	IT	30000	24

TABLE 2: PROJECT

project_id	project_name	budget	duration_months
101	Payroll System	200000	6
102	Attendance App	120000	4
103	AI Analytics	500000	12
104	HR Portal	150000	5
105	Finance Tracker	300000	8
106	Chatbot	180000	6

TABLE 3: WORKS_ON

work_id	emp_id	project_id	hours_per_week
1	1	101	20
2	1	103	15
3	2	104	25
4	3	103	30
5	4	105	20
6	6	102	10

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EXERCISE: SQL OPERATORS (PRACTICE QUESTIONS)

A. ARITHMETIC OPERATORS (+, −, *, /, %)

1. Display employee salary after adding a bonus of 5000.
2. Show employee salary after deducting 2000 as tax.
3. Display double the salary of each employee.
4. Show half of each employee's salary.
5. Display remainder when salary is divided by 3.
6. Display project budget increased by 10%.
7. Display project budget divided equally among 5 teams.
8. Show total monthly budget by dividing budget by duration.
9. Display hours worked per week multiplied by 4 (monthly hours).
10. Display remaining hours when weekly hours are divided by 7.

B. RELATIONAL (COMPARISON) OPERATORS (=, <>, >, <, >=, <=)

1. List employees who belong to the IT department.
2. Display employees whose salary is greater than 40,000.
3. Find employees whose age is less than 30.
4. Display projects with budget equal to 200,000.
5. List projects whose duration is not equal to 6 months.
6. Display employees whose salary is greater than or equal to 50,000.
7. List projects whose duration is less than or equal to 5 months.
8. Display employees whose age is not equal to 32.
9. List employees earning less than 35,000.
10. Display projects with budget greater than 150,000.

C. LOGICAL OPERATORS (AND, OR, NOT)

1. Display employees from IT department earning more than 40,000.
2. List employees who are in HR or FINANCE department.
3. Display employees who are not from IT department.
4. Find employees whose age is greater than 25 and salary less than 50,000.
5. Display projects with budget greater than 200,000 and duration more than 6 months.
6. List employees who belong to IT or whose age is below 27.
7. Display employees who are not in HR department.
8. Find projects with duration less than 6 months or budget less than 150,000.
9. Display employees who are in IT and not earning more than 50,000.
10. List employees whose age is not greater than 30.

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D. SPECIAL OPERATORS (IN, NOT IN, BETWEEN, LIKE, IS NULL, IS NOT NULL)

1. Display employees working in IT or HR departments using IN.
2. List employees not working in IT or FINANCE using NOT IN.
3. Display employees whose salary is between 35,000 and 55,000.
4. List projects with budget between 150,000 and 400,000.
5. Display employee names starting with letter 'A'.
6. List employee names ending with letter 'a'.
7. Display projects whose name contains the word "System".
8. Find employees whose age is between 25 and 30.
9. Display records from WORKS_ON where hours_per_week is NOT NULL.
10. Display employees whose department value is NOT NULL.

E. SET OPERATORS (UNION, UNION ALL, INTERSECT, EXCEPT / MINUS)

1. Display employee IDs from IT department UNION employee IDs from HR department.
2. Display employee IDs from IT department UNION ALL employee IDs from HR department.
3. Display employee IDs who are in IT department and also working on projects.
4. Display employee IDs who are in EMPLOYEE table but not assigned to any project.
5. Display project IDs with budget above 200,000 UNION project IDs with duration above 6 months.
6. Display project IDs common to WORKS_ON and PROJECT tables.
7. Display employee IDs who work on projects EXCEPT those from HR department.
8. Display employee IDs from WORKS_ON INTERSECT employees aged above 30.
9. Display project IDs from PROJECT table EXCEPT those present in WORKS_ON.
10. Display employee IDs from IT UNION employee IDs from WORKS_ON.

F. PATTERN MATCHING OPERATORS (LIKE with % and _)

1. Display employees whose name starts with 'D'.
2. Display employees whose name ends with 'l'.
3. Display employees whose name contains 'ar'.
4. Display projects whose name starts with 'A'.
5. Display projects whose name ends with 'App'.

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6. Display employees whose name has exactly 5 characters.
7. Display employees whose second character is 'h'.
8. Display projects whose name contains the letter 'o'.
9. Display employees whose name does not start with 'E'.
10. Display projects whose name contains exactly one word before 'System'.